PREGNANCY SUPPORT BELT

Electronics Component Selection Report

Preliminary component choices for a wearable engineering prototype exploring movement/orientation sensing. Includes rationale, alternatives, integration notes, and clickable technical/sourcing references.

Recommended prototype	ESP32 development board + GY-521 breakout (MPU-6050) + protected single-cell battery system + removable textile-mounted electronics pod.
Important limitation	This design is not a diagnostic or treatment device and cannot assess fetal health or replace medical advice.
Report basis	Prepared from the project context available in this conversation. The linked shared chat was not extractable here, so confirm exact requirements before purchasing.

Scope and terminology

The commonly referenced sensor is **MPU-6050**; **GY-521** is the name commonly used for a breakout module carrying that sensor. The module is convenient for early tests, but check its actual IC, voltage range, and seller documentation before ordering.

The selections below assume the first prototype records motion/orientation data. If the project needs pressure sensing, app alerts, a specific battery runtime, or a different clinical/research function, the BOM should be revised accordingly.

Prepared: 22 October 2026 | Currency: INR | Pricing and stock must be verified at checkout

1. Recommended component selections

Use development modules for the proof of concept; move to a custom PCB only after sensor placement, firmware, current draw, and mechanical fit have been tested.

Block	Recommended selection	Rationale	Checks / caveats
Controller	**ESP32 DevKit** (ESP32-WROOM-32 class)	Accessible development ecosystem; I²C support and optional Bluetooth/Wi-Fi data transfer.	3.3 V logic. Board variants differ in pinout, USB connector, regulator and battery-input support.
Motion sensor	**GY-521 breakout** using MPU-6050	6-axis accelerometer/gyroscope module for early orientation and movement experiments.	The bare MPU-6050 has lifecycle/availability concerns; verify module IC and source before committing to production.
Battery	**Protected 1-cell 3.7 V LiPo**, initial capacity around 1000 mAh	Compact rechargeable supply for prototype tests; final capacity should follow measured current and runtime needs.	Confirm dimensions, connector polarity, protection, discharge rating and reputable sourcing. Protect from bending, puncture and compression.
Charging	Single-cell charger module with protection, or a board with integrated LiPo charging	Convenient for bench prototyping and reduces custom circuitry in the first iteration.	A TP4056 board is not automatically a complete power-path solution. Verify protection and charge current match the cell. Do not charge while worn.
Regulation	Documented board regulator or suitable 3.3 V regulator	Provides a stable supply to the controller and sensor.	Never connect a raw LiPo to a 3.3 V pin. Check the exact board's supported input path.
Switch	Small on/off switch	Allows power isolation during storage and handling.	Choose suitable current rating; make it accessible without creating a pressure point.
Wiring	Short flexible wires and locking connector with strain relief	Helps prevent disconnections during movement.	Insulate exposed conductors; secure all joints and keep electronics removable for cleaning.
Mounting	Removable padded pocket / enclosure on belt	Maintains repeatable sensor orientation and keeps electronics away from fabric washing.	No hard electronics or battery should press into the abdomen; validate comfort and fit using a non-powered mock-up first.

Recommended first build: ESP32 + GY-521 powered by USB for initial firmware and sensor tests. Add battery and charging after you have measured the current draw and validated sensor data.

2. Alternatives and integration

Subsystem	Economy / quick start	Preferred balanced option	Upgrade path
Controller	Generic ESP32 DevKit	Traceable board with documented pinout and regulator	Compact ESP32 module / custom PCB after validation
Motion sensor	GY-521 / MPU-6050 breakout	Currently supported IMU breakout with datasheet and clear supply range	Low-power IMU selected after accuracy and drift tests
Battery	Protected 1S LiPo around 500–1000 mAh	Size from measured average current and required runtime	Qualified pack with engineered protection and enclosure
Charging	Protected single-cell charger module for bench use	Integrated charger / power-path solution with limits checked	PCB-integrated charging, protection, indicators and test points
Packaging	Secured soft pouch and strain-relieved wires	Washable textile pocket with separate electronics pod	Custom enclosure after fit and cleaning requirements are known

Electrical integration notes

I²C wiring: Connect the GY-521 SDA and SCL to two suitable ESP32 GPIOs, with common ground. GPIO 21 (SDA) and GPIO 22 (SCL) are common defaults on classic ESP32 DevKit boards, but confirm the exact board and firmware configuration. Verify the module's VCC range and pull-up arrangement.

Power path: USB/charger input → single-cell charger/protection → battery → documented board input or designed regulator → controller and sensor. Do not assume every ESP32 board supports a direct battery input, or that USB and battery can safely be connected simultaneously.

Runtime estimate: Approximate runtime in hours = usable battery capacity in mAh ÷ measured average current in mA. For example, 1000 mAh at a measured 100 mA gives a theoretical 10 hours before conversion losses and real-world factors. Measure the prototype under intended sampling and wireless settings.

Sensor capability: An IMU measures acceleration and angular velocity. It does not directly measure belt pressure, fetal movement, fetal wellbeing, or a medical outcome. Any clinical claims need separate validation and appropriate clinical oversight.

> **Safety gate:** Start with USB-powered bench testing. Keep the battery protected and mechanically secured; do not charge while worn. Stop using a damaged, swollen, or unusually hot battery. Do not use the prototype for pregnancy-care decisions.

3. Clickable procurement and technical references

These are starting points for sourcing and design references, not guaranteed stock or price quotes. Check exact part numbers, delivery, tax, and lifecycle status before ordering.

Item	Reference links	Selection note
ESP32 development boards	Espressif hardware / DevKits Official Espressif KiCad libraries	Use a board with a known pinout. Official libraries can help with later schematic/PCB work.
MPU-6050 / GY-521	DigiKey India: MPU-6050 IC listing Find GY-521 breakout modules	The bare IC listing may show obsolete status; verify the exact module and chip before purchasing.
Protected LiPo	Search protected 3.7 V LiPo options	Choose based on dimensions, protection, connector polarity and supplier documentation.
Single-cell charger	Search charger modules with protection	Confirm protection and correct charge current for the selected cell; charger boards differ significantly.
KiCad	KiCad official site Schematic editor documentation	Run electrical rule checks and verify symbols/footprints before PCB fabrication.

4. Suggested build sequence

1. Test ESP32 firmware over USB and confirm serial output.
2. Connect the GY-521 over I²C and log raw accelerometer/gyroscope readings.
3. Calibrate in a fixed orientation and check repeatability before interpreting motion.
4. Measure average current under intended settings and calculate a realistic battery target.
5. Add the selected protected charging/power arrangement and test it on the bench.
6. Build a removable, padded electronics pod; test fit and comfort first with a non-powered mock-up.
7. Only then finalize the KiCad schematic/PCB, footprints, test points, component lifecycle and assembly plan.

Status: preliminary engineering recommendation. Confirm sensing requirements, desired runtime, available space, and whether any clinical/research use is intended before freezing the BOM.